

GUGA, Russian Federation

WMO: 314210

Lat: **52.700N**Lon: **137.533E**Elev: **60**StdP: **100.61**Time Zone: **10.00 (E10)**Period: **82-94**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.0	-38.8	-43.4	0.1	-40.7	-41.3	0.1	-38.6	4.1	-17.2	3.4	-18.2	0.0	N/A	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	30.9	20.5	28.9	19.5	26.8	18.5	21.7	28.1	20.7	27.0	19.6	25.2	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.5	24.5	18.5	13.5	23.1	17.4	12.5	22.2	64.0	28.2	60.2	27.4	56.3	25.6	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.1	4.1	3.3	-43.3	34.6	2.5	2.4	-45.1	36.3	-46.5	37.7	-47.9	39.0	-49.7	40.8		
			-42.9	25.1	2.4	2.7	-44.6	27.0	-46.1	28.6	-47.4	30.1	-49.2	32.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-26.8	-19.9	-11.3	0.5	7.9	14.7	18.6	16.6	9.8	0.2	-13.9	-24.4
	DBStd	16.52	5.99	6.04	5.97	4.32	4.36	4.32	3.60	2.90	3.76	4.80	6.38	6.29
	HDD10.0	5167	1142	837	659	286	96	9	0	0	48	305	718	1067
	HDD18.3	7595	1400	1071	918	536	323	123	43	69	256	563	968	1325
	CDD10.0	696	0	0	0	1	31	151	266	206	43	0	0	0
	CDD18.3	82	0	0	0	0	0	14	51	17	0	0	0	0
	CDH23.3	1330	0	0	0	0	49	363	674	238	6	0	0	0
	CDH26.7	430	0	0	0	0	7	121	239	62	1	0	0	0
Wind	WSAvg	0.9	0.4	0.6	0.8	1.4	1.4	1.3	1.0	0.8	0.8	0.9	0.7	0.3
Precipitation	PrecAvg	724	48	28	38	39	64	61	74	90	76	77	71	58
	PrecMax	949	100	72	92	92	125	127	147	187	139	159	150	115
	PrecMin	517	1	1	5	17	5	12	8	18	25	23	25	5
	PrecStd	98	29	20	25	15	28	26	37	38	28	34	33	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.5	-2.4	9.7	18.2	27.5	32.9	33.1	32.0	24.1	16.3	2.6	-2.7
		MCWB	-5.5	-4.7	3.3	8.8	15.3	21.1	20.9	21.2	17.3	8.9	0.2	-4.1
	2%	DB	-9.0	-4.9	5.4	14.2	24.4	30.3	31.3	28.3	21.4	13.6	0.3	-8.0
		MCWB	-10.0	-6.5	0.3	6.2	13.7	19.6	20.5	19.7	15.3	7.4	-1.7	-9.1
	5%	DB	-11.7	-6.9	2.2	11.5	21.1	27.9	29.6	26.0	18.6	10.3	-1.3	-10.8
		MCWB	-12.6	-8.4	-1.4	4.7	12.2	17.8	20.2	18.5	14.0	5.4	-3.2	-11.6
	10%	DB	-17.2	-9.7	-0.4	8.8	17.6	25.1	27.6	23.8	16.5	7.9	-3.3	-13.9
		MCWB	-17.9	-11.1	-3.2	3.1	10.6	16.8	19.9	17.9	12.8	4.5	-4.9	-14.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.6	-4.4	3.7	9.1	16.3	22.7	23.5	23.2	18.1	9.7	1.1	-4.1
		MCDB	-4.5	-2.7	8.3	17.2	25.0	30.7	28.6	29.1	22.5	14.4	1.8	-2.8
	2%	WB	-9.9	-6.3	0.7	6.7	14.4	20.6	22.1	21.0	16.4	7.6	-1.3	-9.0
		MCDB	-9.0	-4.9	4.7	13.3	22.7	29.0	28.1	26.8	19.6	11.9	0.2	-8.1
	5%	WB	-12.7	-8.4	-1.2	5.0	12.8	18.9	21.1	19.7	15.1	6.3	-3.3	-11.6
		MCDB	-11.7	-6.9	2.1	10.4	20.1	25.9	27.4	24.1	17.9	10.1	-1.6	-10.9
	10%	WB	-17.8	-11.0	-3.2	3.5	10.9	17.3	20.2	18.5	13.6	4.5	-4.9	-14.6
		MCDB	-17.3	-9.7	-0.6	8.2	16.9	23.8	26.2	22.3	16.0	7.7	-3.3	-14.0
Mean Daily Temperature Range	5% DB	MDBR	14.9	17.2	17.9	12.9	13.8	15.3	13.4	12.4	11.9	12.2	13.4	12.7
		MCDBR	11.2	17.2	19.1	18.4	21.5	21.1	18.3	18.2	16.0	16.6	12.5	11.3
		MCWBR	10.2	15.0	14.3	10.9	11.5	10.8	8.6	9.9	9.5	10.5	10.1	10.3
	5% WB	MCDBR	10.7	15.6	17.4	17.3	19.3	17.7	15.1	14.7	13.3	15.5	10.9	10.4
		MCWBR	9.7	13.5	12.7	10.5	10.6	9.7	7.8	8.8	8.2	10.1	9.1	9.7
Clear-Sky Solar Irradiance	taub	0.274	0.308	0.379	0.494	0.570	0.537	0.551	0.458	0.388	0.390	0.326	0.278	
	taud	2.066	2.009	1.852	1.701	1.732	1.907	1.908	2.183	2.322	2.106	2.043	2.025	
	Ebn at Noon	684	771	775	727	696	730	710	765	783	665	593	588	
	Edh at Noon	75	109	159	211	216	183	180	130	100	100	78	65	
All-Sky Solar Radiation	RadAvg	1.31	2.48	3.99	5.02	5.03	5.63	5.01	4.26	3.37	2.09	1.28	0.98	
	RadStd	0.07	0.10	0.24	0.31	0.52	0.53	0.56	0.47	0.29	0.18	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page